

PHYSICS SYLLABUS - Class 11, JKBOSE

Max Marks: 100
Theory: 70 Marks

Time Allowed: 3 hrs
Practical: 30 Marks

Unit 1: Physical World and Measurement (05 Marks)

Units and Measurements

- Need for measurement
- Units of measurement
- Systems of units
- SI units
- Fundamental and derived units
- Significant figures
- Dimensions of physical quantities
- Dimensional analysis and its applications

Unit 2: Kinematics (09 Marks)

Motion in a Straight Line

- Frame of reference
- Motion in a straight line
- Elementary concepts of differentiation and integration for describing motion
- Uniform and non-uniform motion
- Instantaneous velocity
- Uniformly accelerated motion
- Velocity–time and position–time graphs
- Relations for uniformly accelerated motion (graphical treatment)

Motion in a Plane

- Scalar and vector quantities
- Position and displacement vectors
- Vector notation and equality of vectors
- Multiplication of vectors by a real number
- Addition and subtraction of vectors
- Unit vector
- Resolution of vectors in a plane
- Rectangular components
- Scalar and vector product of vectors
- Projectile motion
- Uniform circular motion

Unit 3: Laws of Motion (07 Marks)

- Intuitive concept of force and inertia
- Newton's First Law
- Momentum and Newton's Second Law
- Impulse
- Newton's Third Law
- Law of conservation of linear momentum and applications
- Equilibrium of concurrent forces
- Static and kinetic friction
- Laws of friction
- Rolling friction and lubrication
- Dynamics of uniform circular motion
- Centripetal force
- Examples of circular motion (vehicles on level and banked roads)

Unit 4: Work, Energy and Power (06 Marks)

- Work done by a constant and variable force
- Kinetic energy
- Work-energy theorem
- Power
- Potential energy
- Potential energy of a spring
- Conservative and non-conservative forces
- Motion in a vertical circle
- Elastic and inelastic collisions in one and two dimensions

Unit 5: Motion of System of Particles and Rigid Body (06 Marks)

System of Particles

- Centre of mass of a two-particle system
- Momentum conservation
- Centre of mass motion
- Centre of mass of rigid body
- Centre of mass of a uniform rod

Rotational Motion

- Moment of force (torque)
- Angular momentum
- Conservation of angular momentum
- Equilibrium of rigid bodies
- Equations of rotational motion
- Comparison of linear and rotational motion
- Moment of inertia
- Radius of gyration
- Values of moments of inertia for simple geometrical objects (no derivation)

Unit 6: Gravitation (06 Marks)

- Kepler's laws of planetary motion
- Universal law of gravitation
- Acceleration due to gravity and its variation with altitude and depth
- Gravitational potential energy
- Gravitational potential
- Escape speed
- Orbital velocity of a satellite

Unit 7: Properties of Bulk Matter (09 Marks)

Mechanical Properties of Solids

- Elasticity
- Stress-strain relation
- Hooke's law
- Young's modulus
- Bulk modulus
- Shear modulus
- Poisson's ratio
- Elastic energy

Mechanical Properties of Fluids

- Pressure due to fluid column
- Pascal's law and applications
- Hydraulic lift and hydraulic brakes
- Effect of gravity on fluid pressure
- Viscosity
- Stokes' law
- Terminal velocity
- Streamline and turbulent flow
- Critical velocity

- Bernoulli's theorem and applications
- Surface tension and surface energy
- Angle of contact
- Excess pressure across a curved surface
- Applications of surface tension (drops, bubbles, capillary rise)

Thermal Properties of Matter

- Heat and temperature
- Thermal expansion of solids, liquids and gases
- Anomalous expansion of water
- Specific heat capacity
- C_p and C_v
- Calorimetry
- Latent heat
- Heat transfer: conduction, convection and radiation
- Thermal conductivity
- Blackbody radiation
- Wien's displacement law
- Stefan's law

Unit 8: Thermodynamics (06 Marks)

- Thermal equilibrium
- Zeroth law of thermodynamics
- Heat, work and internal energy
- First law of thermodynamics
- Second law of thermodynamics
- Isothermal and adiabatic processes
- Reversible and irreversible processes
- Cyclic processes

Unit 9: Behaviour of Perfect Gases and Kinetic Theory (06 Marks)

- Equation of state of a perfect gas
- Work done in compressing a gas
- Assumptions of kinetic theory
- Pressure exerted by a gas
- Kinetic interpretation of temperature
- RMS speed of gas molecules
- Degrees of freedom
- Law of equipartition of energy (statement only)
- Specific heat capacities of gases
- Mean free path
- Avogadro's number

Unit 10: Oscillations and Waves (10 Marks)

Oscillations

- Periodic motion
- Time period and frequency
- Displacement as a function of time
- Periodic functions
- Simple harmonic motion and its equation
- Phase
- Oscillations of a loaded spring
- Restoring force and force constant
- Energy in SHM
- Simple pendulum and expression for its time period

Waves

- Transverse and longitudinal waves
- Speed of travelling wave
- Displacement relation for a progressive wave
- Principle of superposition
- Reflection of waves
- Standing waves in strings and organ pipes
- Fundamental mode and harmonics
- Beats